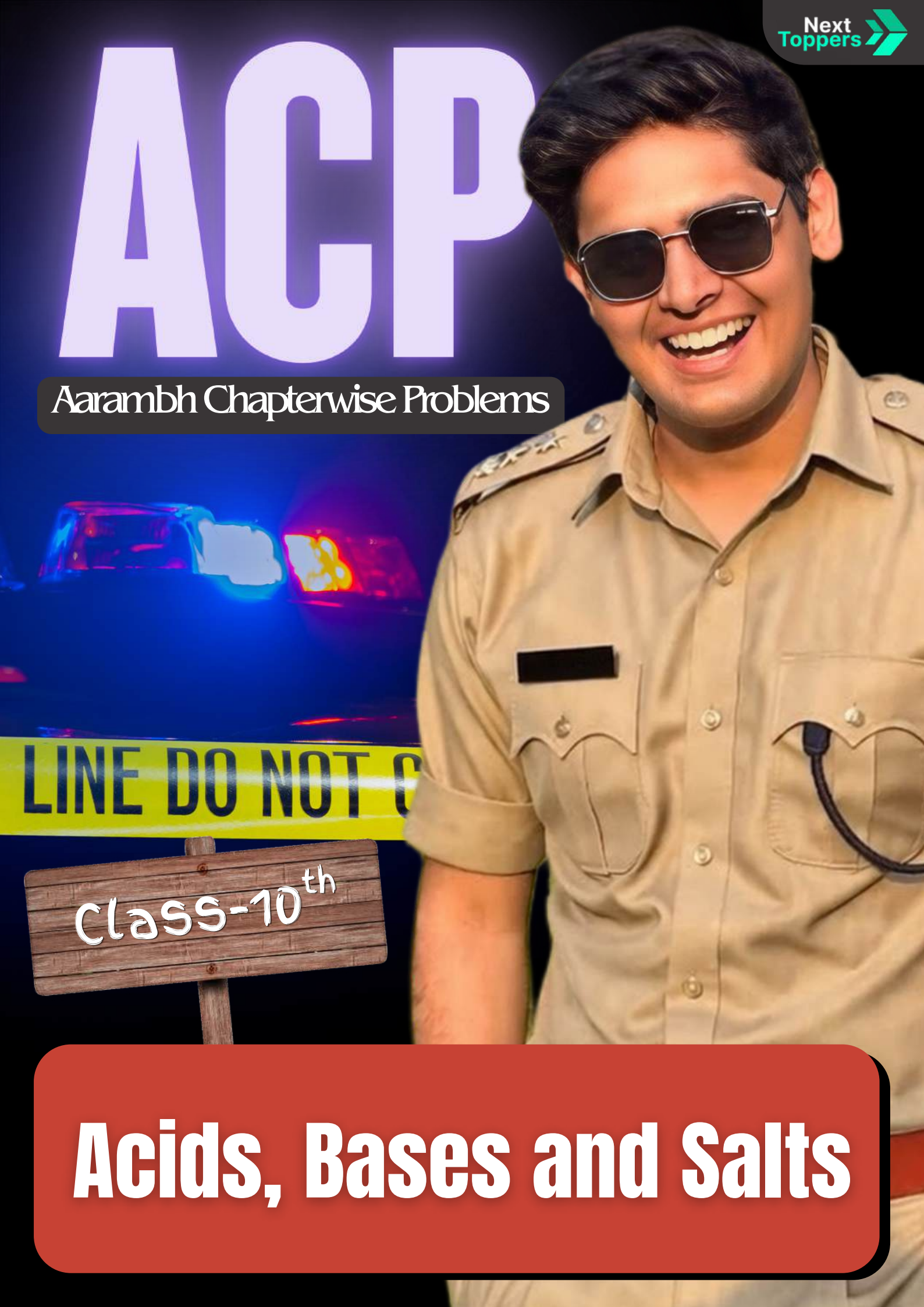


ACP

Aarambh Chapterwise Problems



LINE DO NOT CROSS

Class-10th

Acids, Bases and Salts

Acids, Bases & Salts

Read the question properly!

PART 1 (Basic MCQs)

1. Farmers neutralise the effect of Acidity on the soil by adding –

- (a) Slaked lime
- (b) Gypsum
- (c) Caustic soda
- (d) Baking soda

2. An aqueous solution turns the red litmus solution blue. Excess addition of which of the following solutions would reverse the change?

- (a) Baking powder
- (b) Lime
- (c) Ammonium hydroxide solution
- (d) Hydrochloric acid

3. In which pH range does our body work to survive in the atmosphere?

- (a) 5.5 to 8.5
- (b) 7.0 to 7.8
- (c) 2.3 to 7.0
- (d) 7.5 to 12.5

4. What happens when a solution of an acid is mixed with a solution of a base in a test tube?

- (i) The temperature of the solution increases
- (ii) The temperature of the solution decreases
- (iii) The temperature of the solution remains the same
- (iv) Salt formation takes place

- (a) (i) only
- (b) (i) and (iii)
- (c) (ii) and (iii)
- (d) (i) and (iv)

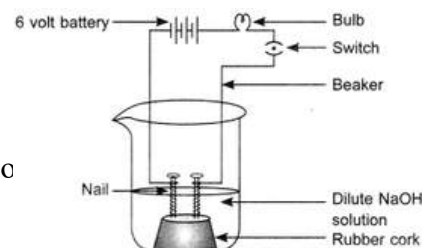
5. The apparatus given in the adjoining figure was set up to demonstrate electrical conductivity

Which of the following statement(s) is (are) correct?

- (i) Bulb will not glow because NaOH solution is not acidic.
- (ii) Bulb will glow because NaOH is a strong base and furnishes ions for conduction.
- (iii) Bulb will not glow because the circuit is incomplete.
- (iv) Bulb will glow because NaOH solution conducts electricity due to the presence of ions.

- (a) (i) and (iii)
- (b) (ii) and (iv)
- (c) (ii) only
- (d) (iv) only

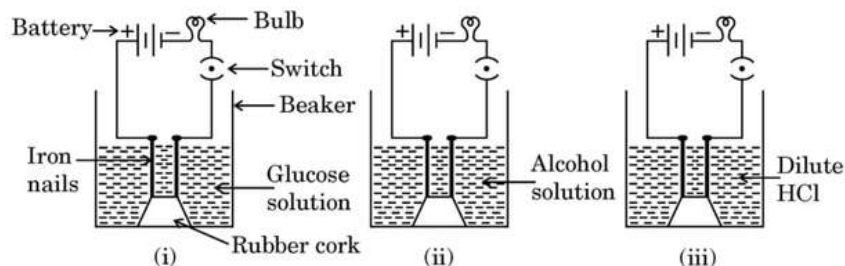
**JOSH
METER**



6. When a base reacts with a metal, it forms a salt, and hydrogen gas is released. By what method can the presence of hydrogen be detected?

- (a) by water
- (b) by litmus paper
- (c) by methyl orange
- (d) by a burning candle

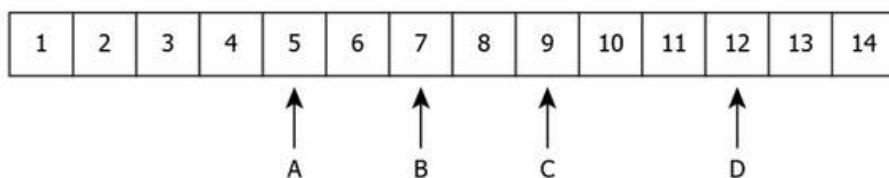
7. In which case/cases will the bulb glow considering that all other connections of electric circuit are complete and working?



- (a) Case (i) only
- (b) Case (ii) only
- (c) Cases (ii) and (iii)
- (d) Case (iii) only

8. The image shows the pH values of four solutions on a pH scale.

Which solutions are alkaline in nature?

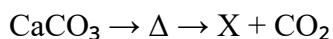


- (a) A and B
- (b) B and C
- (c) C and D
- (d) A and D

9. How will you protect yourself from the heat generated while diluting a concentrated acid?

- (a) By adding acid to water with constant stirring.
- (b) By adding water to acid with constant stirring.
- (c) By adding water to acid followed by base.
- (d) By adding base to acid with constant stirring

10. Identify the product 'X' obtained in the following chemical reaction:

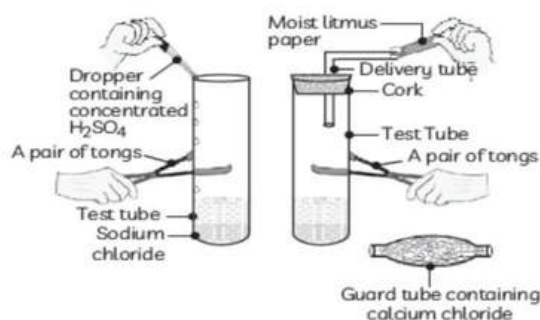


- (a) Quick lime
- (b) Gypsum
- (c) Lime stone
- (d) Plaster of Paris

11. The change in the colour of the moist litmus paper in the given set up is due to:

- (i) presence of acid
- (ii) presence of base
- (iii) presence of $\text{H}^+(\text{aq})$ in the solution
- (iv) presence of litmus which acts as an indicator

- (a) (i) and (ii)
- (b) Only (ii)
- (c) Only (iii)
- (d) Only (iv)



12. The warning sign shown in the given figure must invariably be displayed/pasted on the containers which contain hydroxide of:

- (a) Aluminium
- (b) Calcium
- (c) Sodium
- (d) Magnesium



13. A few pieces of granulated zinc are taken in a test tube and 2 mL of sodium hydroxide solution is added to it. When the contents are warmed, the product formed is:

- (a) Na_2ZnO
- (b) NaZnO_2
- (c) Na_2ZnO_2
- (d) $\text{NaZn}(\text{OH})_2$

14. A common feature observed in the crystals of washing soda, copper sulphate, gypsum and ferrous sulphate is that all

- (a) exhibit basic nature
- (b) exhibit acidic nature
- (c) have a fixed number of water molecules of crystallisation
- (d) are coloured

15. When a mixture of baking soda and tartaric acid is heated (or mixed in water) a product 'X' is formed, which is responsible for making breads and cakes soft and spongy. The product 'X' is

- (a) Carbon dioxide
- (b) Carbon monoxide
- (c) Sodium tartrate
- (d) Hydrogen

16. You have three aqueous solutions A, B and C as given below :

- (a) Potassium nitrate
 - (b) Ammonium chloride
 - (c) Sodium carbonate Choose the correct increasing order of pH is:
- (a) $A < B < C$
 - (b) $B < C < A$
 - (c) $C < A < B$
 - (d) $B < A < C$

17. Solid Calcium oxide reacts vigorously with water to form Calcium hydroxide accompanied by the liberation of heat. From the information given above it may be concluded that this reaction

- (a) is endothermic and pH of the solution formed is more than 7.
- (b) is exothermic and pH of the solution formed is 7.
- (c) is endothermic and pH of the solution formed is 7.
- (d) is exothermic and pH of the solution formed is more than 7.

ASSERTION AND REASON TYPE QUESTION (1 Marks)*iyaha marks katate hi*

In the following questions a statement of Assertion is followed by a statement of Reason.

Mark the correct choice as two statements are given one labeled Assertion (A) and the other labeled Reason (R).

Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below:

- (a) Both A and R are true, and R is correct explanation of the assertion.
- (b) Both A and R are true, but R is not the correct explanation of the assertion.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

18. Assertion: Solutions of compounds like alcohol and glucose do not show acidic character.

Reason: They do not show acidic character because they do not dissociate into ions.

19. Assertion: Dry HCl gas does not change the colour of the dry litmus paper.

Reason: It is because dry HCl does not contain the H^+ ions.

20. Assertion: An aqueous solution of acid conducts electricity.

Reason: It is because in the solution it dissociates into ions.

21. Assertion: Tap water conducts electricity but distilled water does not conduct electricity.

Reason: Tap water conducts electricity as it contains ions whereas distilled water does not contain ions.

22. Assertion: Curd and sour substances should not be stored in copper vessels.

Reason: Curd and other sour substances should not be kept in brass and copper vessels as they contain acids.

23. Assertion: Plaster of Paris should be stored in dry place.

Reason: To prevent its reaction with moisture.

24. Assertion (A): Fresh milk in which baking soda is added, takes a longer time to set as curd.

Reason (R): Baking soda decreases the pH value of fresh milk to below 6.

*tho jaayenge aaram se***PART 2 (Subjective Questions)****SHORT ANSWER TYPE QUESTIONS (2 and 3 Marks)**

25. A cloth strip dipped in onion juice is used for testing a liquid 'X'. The liquid 'X' does not change the odour of onion juice. Which type of indicator is onion juice?

The liquid 'X' turns blue litmus red. List the observations the liquid 'X' will show on reacting with the following:

- (a) Zinc granules
- (b) Solid sodium carbonate

Write the chemical equations for the reactions involved.

26. A student prepared solutions of (i) an acid and (ii) a base in two separate beakers. She forgot to label the solutions, and litmus paper was not available in the laboratory. Since both the solutions are colourless, how will she distinguish between the two?

27. In one of the industrial processes used to manufacture sodium hydroxide, a gas X is formed as a byproduct. The Gas X reacts with dry slaked lime $[\text{Ca}(\text{OH})_2]$ to form compound Y used as a bleaching agent in the chemical industry. Identify X and Y giving the chemical equation of the reactions involved.
28. When zinc metal is treated with a dilute solution of a strong acid, a gas is evolved, which is utilised in the hydrogenation of oil. Name the gas evolved. Write the chemical equation of the reaction involved and also write a test to detect the gas formed.
29. (a) A compound 'A' is used in a soda-acid fire extinguisher and decomposes on heating to form compound 'B'. Compound 'B' is used for removing permanent hardness of water. Identify 'A' and 'B'.
(b) Write the chemical formula of Plaster of Paris. At what temperature is gypsum heated to obtain Plaster of Paris?
30. Explain chlor-alkali process with chemical equation. Name the products formed at anode and cathode.
31. (i) Under what condition is rainwater called acid rain?
(ii) In the following acids, identify weak acids:
Nitric acid, hydrochloric acid, acetic acid, sulphuric acid, formic acid.

LONG ANSWER TYPE QUESTIONS (5 Marks)

(pahle points socho fir likho)

32. In the following schematic diagram for the preparation of hydrogen gas as shown in Figure 2.3, what would happen if following changes are made?

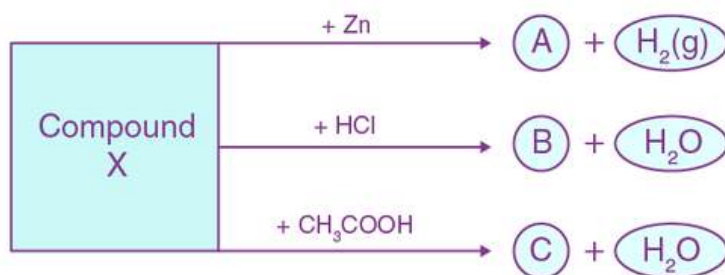


- (a) In place of zinc granules, the same amount of zinc dust is taken in the test tube
(b) Instead of dilute sulphuric acid, dilute hydrochloric acid is taken
(c) In place of zinc, copper turnings are taken
(d) Sodium hydroxide is taken in place of dilute sulphuric acid and the tube is heated.
33. For making cake, baking powder is taken. If your mother uses baking soda instead of baking powder in cake at home,
(a) How will it affect the taste of the cake and why?
(b) How can baking soda be converted into baking powder?
(c) What is the role of tartaric acid added to baking soda?
34. A metal carbonate X reacting with acid gives a gas that gives the carbonate back when passed through a solution Y. On the other hand, a gas G obtained at the anode during electrolysis of brine is passed on dry Y, it gives a compound Z, used for disinfecting drinking water. Identity X, Y, and Z.

35. A dry pellet of a common base B absorbs moisture and turns sticky when kept open. The compound is also a product of the chlor-alkali process. Identify B. What type of reaction occurs when B is treated with an acidic oxide? Write a balanced chemical equation for one such solution.

36. A sulphate salt of Group 2 element of the Periodic Table is a white, soft substance, which can be moulded into different shapes by making its dough. When this compound is left open for some time, it becomes a solid mass and cannot be used for moulding purposes. Identify the sulphate salt and why does it show such behaviour? Give the reaction involved.

37. Identify the compound X based on the reactions given below. Also, write the name and chemical formulae of A, B and C.



10th
Phodenge!



38. (i) Identify the acid and the base from which NaCl is obtained. Which type of salt is it? When is it called rock salt? How is rock salt formed?

(ii) Blue litmus solution is added to two test tubes A and B containing dilute HCl and NaOH solution respectively. In which test tube, a colour change will be observed? State the colour change and give its reason.

39. The pH plays an important role in our daily life. Strength of acids and bases depends on the number of H^+ ions and OH^- ions produced respectively. pH scale helps in predicting the strength of acids and bases by measuring hydrogen ion concentration in a solution.

Answer the following:

(a) What type of indicator is used to predict how strong an acid or base is?

(b) Three solutions A, B and C showed pH as 3, 5 and 7. Which solution has the highest concentration of H^+ ions?

(c) What are strong acids and weak acids?

40. Seawater contains many salts dissolved in it. Common salt is separated from these salts. Deposits of solid salt are also found in several parts of the world. These large crystals are often brown due to impurities. This is called rock salt and is mined like coal. The common salt is an important raw material for chemicals of daily use. Write balanced chemical equations showing the products formed during electrolysis of brine. List two uses of any one of the products.

OR

Common salt is a very important chemical compound for our daily life. Its chemical name is sodium chloride and it is used as a raw material in the manufacture of caustic soda, washing soda, baking soda etc. It is also used in the preservation of pickles, butter, meat etc.

(i) Name the acid and the base from which common salt can be obtained.

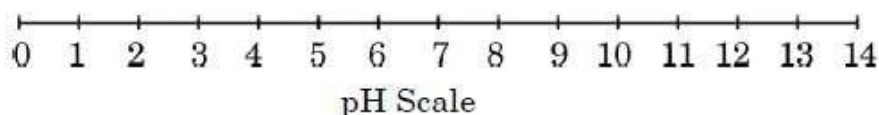
(ii) State the nature (acidic/basic/neutral) of sodium chloride. Give reason for the justification for your answer.

(iii) (a) What happens when electric current is passed through an aqueous solution of sodium chloride (called brine)? Name the products obtained along with the corresponding places in the electrolytic cell where each of these products is obtained.

PART 3 (Advanced Questions)

Case Study/Source Based Question (5 Marks)

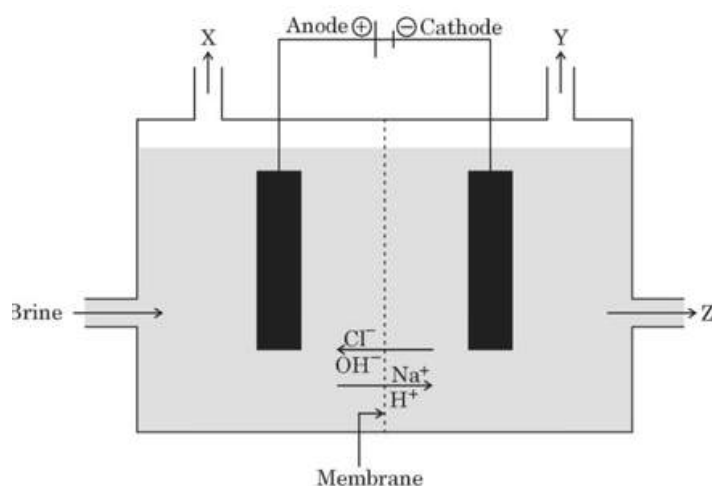
41. The following questions are source-based/case-based questions. Read the case carefully and answer the questions that follow. Acid-base indicators can be used to distinguish between an acid and a base. Universal indicator, a mixture of several indicators, shows different colours at different concentrations of acids and bases, thereby indicating their pH on the pH scale of 0 - 14. The pH of a solution is measured by pH paper, which is a paper impregnated with a universal indicator.



Answer the following questions:

- (a) Solution P is a strong acid, while solution Q is a strong base. On the pH scale, where would you place the solutions P and Q ?
- (b) A solution has a pH of 7. Name a compound you would use to
- (i) Increase its pH (ii) Decrease its pH
- (c) (i) When the pH of a solution is decreased from 4 to 2, what effect does it produce on its hydronium ion concentration ? State the colour change shown by the pH paper.

42. Read the following passage and answer the questions that follow : In the given diagram, when electricity is passed through an aqueous solution of a common salt, a substance 'Z' is produced, along with the evolution of gases 'X' and 'Y'. When a burning matchstick is brought near the gas 'Y', it burns with a pop sound, whereas 'X' is used for disinfecting drinking water. When 'X' is passed through slaked lime, an insoluble substance 'A' is produced.

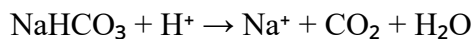


10th
Phodenge!



- (a) Write the names of the gases 'X' and 'Y'.
- (b). Write the balanced chemical equation for the formation of substance 'Z'.
- (c) (i). What happens when:
- (I) If a drop of red litmus solution is added to the aqueous solution of substance 'Z'?
- (II) If CO₂ gas is passed through ammoniacal solution of aqueous NaCl?
- (III). Identify the substance 'A' and write the balanced chemical equation of the reaction which takes place in its preparation.

43. Baking powder is used in making cakes, bread and pastries to make them soft and spongy. It is used in pakoda, but is not directly used; instead, baking soda is used. Baking powder contains sodium hydrogencarbonate and a mild edible acid. When baking powder mixes with water, the sodium hydrogencarbonate reacts with the acid to evolve carbon dioxide gas which gets trapped in the wet dough and bubbles out slowly making the cake rise and become 'soft and spongy' thus imparting the mixture with a light, fluffy texture. The chemical reaction which takes place can be shown as:



Questions:

1. Sodium hydrogencarbonate is prepared by:
 - (a) Sodium chloride + ammonia + water + carbon dioxide
 - (b) Sodium chloride + ammonia
 - (c) Ammonia + carbon dioxide
 - (d) All of these
2. The gas released during heating of sodium hydrogencarbonate is:
 - (a) Sulphur dioxide
 - (b) Hydrogen
 - (c) Carbon dioxide
 - (d) Carbon monoxide
3. The edible acid added to the sodium hydrogencarbonate to make baking powder is:
 - (a) Tartaric acid
 - (b) Tartaric acid + hydrochloric acid
 - (c) Formic acid
 - (d) Nitric acid
4. Sodium hydrogencarbonate is used to prepare cakes because:
 - (a) It makes the cake tasty
 - (b) It makes the cake sweet
 - (c) It makes the cake soft and spongy
 - (d) All of these

44. Riya noticed that her grandmother used a white powder to clean greasy utensils. The powder was also used in baking cakes. During a science activity, Riya added a few drops of dilute hydrochloric acid to this powder and observed brisk effervescence. The gas produced turned lime water milky.

Questions

1. The white powder is most likely:
 - (a) Washing soda
 - (b) Baking soda
 - (c) Plaster of Paris
 - (d) Bleaching powder
2. The gas evolved is:
 - (a) Oxygen
 - (b) Hydrogen
 - (c) Carbon dioxide
 - (d) Chlorine
3. Which statement is correct?
 - (a) The powder is acidic in nature.
 - (b) The powder neutralizes acids completely without producing gas.
 - (c) The powder is a salt obtained from a strong acid and weak base.
 - (d) The powder can be used as an antacid.

10th
Phodenge!



4. If the same powder is heated strongly, the products formed are:

- (a) NaOH , CO_2 and H_2O
- (b) Na_2CO_3 , CO_2 and H_2O
- (c) NaCl and H_2O
- (d) Na_2O and CO_2

45. A school laboratory accidentally spilled a strong acid on a marble floor. After a few days, white patches appeared and the floor surface became rough. To investigate, students collected a small marble sample (calcium carbonate) and added dilute hydrochloric acid. A gas was evolved.

Questions

1. Marble mainly contains:

- (a) Calcium oxide
- (b) Calcium hydroxide
- (c) Calcium carbonate
- (d) Calcium chloride

2. The roughening of the floor occurred because:

- (a) Acid reacted with calcium carbonate.
- (b) Acid reacted with oxygen in air.
- (c) Acid evaporated and left deposits.
- (d) Acid converted marble into a metal.

3. The gas evolved in the reaction would:

- (a) Rekindle a glowing splint
- (b) Burn with a pop sound
- (c) Turn lime water milky
- (d) Bleach moist litmus paper

4. If the acid used was sulfuric acid instead of hydrochloric acid, which salt would be formed?

- (a) Calcium chloride
- (b) Calcium sulfate
- (c) Calcium sulfide
- (d) Calcium sulfite

46. A municipal water treatment plant disinfects drinking water before supplying it to homes. During one inspection, students observed that a white powder with a strong smell was being added to water tanks. The engineer explained that this substance is prepared by passing chlorine gas over dry slaked lime.

Later, the students learned that the same substance gradually loses its effectiveness when exposed to moist air for a long time.

Questions

1. Identify the white powder being used in the treatment plant.

2. Write the chemical reaction involved in its preparation.

3. Why is this substance added to drinking water?

4. Why should this substance be stored in airtight containers?

5. A student suggests using baking soda instead of this substance for water purification. Do you agree? Give reasons.

6. Mention one other practical application of this substance besides water treatment.

10th
Phodenge!



- Acids and Bases - Q1, Q2, Q4, Q9, Q11, Q17, Q18, Q19, Q20, Q22, Q24, Q25, Q26, Q28, Q31, Q32
- pH Scale and Indicators - Q3, Q8, Q16, Q39, Q41
- Conductivity of Acid/Base Solutions - Q5, Q7, Q21
- Reaction of Acids with Metals - Q6, Q28, Q32
- Reaction of Acids with Carbonates and Bicarbonates - Q10, Q25, Q45
- Water of Crystallisation - Q14
- Salts and Their Properties - Q16, Q38
- Sodium Zincate - Q13, Q37
- Common Salt and Chlor-Alkali Process - Q27, Q30, Q40, Q42
- Sodium Hydroxide (Caustic Soda) - Q12, Q35, Q37
- Bleaching Powder - Q34, Q42, Q46
- Baking Soda and Baking Powder - Q15, Q29, Q33, Q43, Q44
- Plaster of Paris and Gypsum - Q23, Q29, Q36
- Acid Rain - Q31
- Neutralisation Reactions - Q4, Q17, Q35, Q37, Q38
- Everyday Life Applications of Acids, Bases and Salts - Q1, Q3, Q15, Q22, Q33, Q44, Q46
- Weak and Strong Acids - Q31, Q39



SOLUTIONS

MULTIPLE CHOICE QUESTIONS [MCQ's]

- | | | |
|--------|--------|--------|
| 1.(a) | 11.(c) | 21.(a) |
| 2.(d) | 12.(c) | 22.(a) |
| 3.(b) | 13.(c) | 23.(a) |
| 4.(d) | 14.(c) | 24.(c) |
| 5.(b) | 15.(a) | |
| 6.(d) | 16.(d) | |
| 7.(d) | 17.(d) | |
| 8.(c) | 18.(a) | |
| 9.(a) | 19.(a) | |
| 10.(a) | 20.(a) | |

SHORT ANSWER TYPE QUESTIONS:

25. Onion juice is an olfactory indicator because its smell changes in basic solutions and remains unchanged in acidic solutions.

Since liquid X turns blue litmus red, liquid X is an acid.

Observation with zinc granules: Hydrogen gas is evolved with brisk effervescence.

Equation: $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\uparrow$

Observation with solid sodium carbonate:

Carbon dioxide gas is evolved with brisk effervescence. This gas turns lime water milky.

Equation: $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2\uparrow$

26. Use a natural indicator such as china rose (hibiscus) indicator.

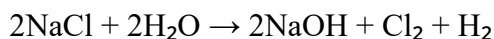
- Acid $\rightarrow$ dark pink/magenta colour
- Base $\rightarrow$ green colour

Thus the acid and base can be distinguished.

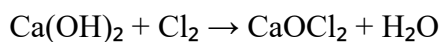
27. X = Chlorine gas (Cl_2)

Y = Bleaching powder (CaOCl_2)

Electrolysis of brine (Chlor-alkali process):



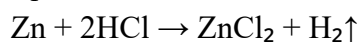
Formation of bleaching powder:



Thus, chlorine gas produced during the chlor-alkali process reacts with dry slaked lime to form bleaching powder, which is used as a bleaching and disinfecting agent.

28. Gas evolved: Hydrogen (H_2)

Equation:

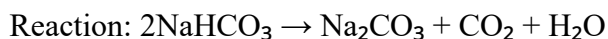


Test: Hydrogen burns with a pop sound when a burning splint is brought near it.

29. (a)

A = Sodium hydrogen carbonate (NaHCO_3)

B = Sodium carbonate (Na_2CO_3)



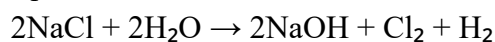
(b) Plaster of Paris: $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$

Temperature: 373–393 K (100–120°C)

30. Chlor-alkali Process

Electrolysis of brine produces sodium hydroxide, chlorine and hydrogen.

Equation:



Anode: Chlorine gas (Cl_2)

Cathode: Hydrogen gas (H_2)

31.

(i)

Rainwater is called acid rain when its pH is less than 5.6.

(ii)

Weak acids:

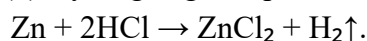
- Acetic acid
- Formic acid

LONG ANSWER TYPE QUESTIONS:

32.

(a) Zinc dust reacts faster than zinc granules because it has a larger surface area.

(b) Hydrogen gas is produced.



(c) Copper does not react with dilute sulphuric acid. Hence no hydrogen gas is produced.

(d) Hydrogen gas will be evolved and sodium zincate will be formed.

33.

(a) Cake will taste bitter because baking soda on heating forms sodium carbonate.

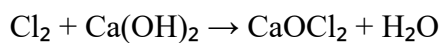
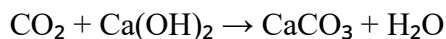
(b) Baking powder is prepared by mixing baking soda with tartaric acid.

(c) Tartaric acid neutralises sodium carbonate formed during heating and prevents bitterness.

34.

X = Calcium carbonate (CaCO_3)Y = Slaked lime, $\text{Ca}(\text{OH})_2$ (dry)Z = Bleaching powder [CaOCl_2]

Explanation:



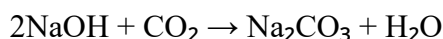
35.

B = Sodium hydroxide (NaOH)

It absorbs moisture from air and becomes sticky (deliquescent).

When treated with an acidic oxide, a neutralisation reaction occurs.

Example:

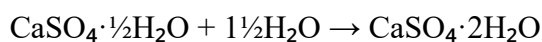


36.

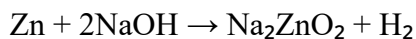
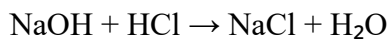
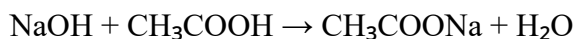
The sulphate salt is Plaster of Paris ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$).

It absorbs moisture from air and changes into gypsum, becoming hard and unusable for moulding.

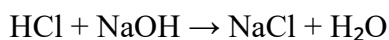
Reaction:



37.

Compound X = Sodium hydroxide (NaOH)A = Sodium zincate (Na_2ZnO_2)B = Sodium chloride (NaCl)C = Sodium acetate (CH_3COONa)

38.

(i) NaCl is obtained from Hydrochloric acid (HCl) and Sodium hydroxide (NaOH). NaCl is a neutral salt.

Rock salt is the naturally occurring impure form of common salt found underground and mined like coal.

(ii) Blue litmus shows no colour change in HCl (remains blue only if dry; in solution it turns red) and remains blue in NaOH . Since the question asks for observation, A (HCl) turns blue litmus red, while B (NaOH) shows no change.

39.

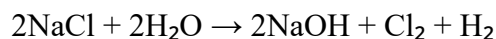
(a) The pH scale is used to predict the strength of acids and bases.

(b) Solution A ($\text{pH} = 3$) has the highest concentration of H^+ ions.

(c)

- Strong acids completely ionise in water (e.g., HCl , H_2SO_4).
- Weak acids partially ionise in water (e.g., CH_3COOH).

40. Electrolysis of brine:



Products formed:

- Chlorine gas at the anode
- Hydrogen gas at the cathode
- Sodium hydroxide solution in the cell

Uses of chlorine:

- Disinfection of drinking water.
- Manufacture of bleaching powder.

CASE STUDY/SOURCE BASED QUESTION:

41. (a) P (strong acid): pH near 1

- Q (strong base): pH near 14

(b) (i) To increase pH: Sodium hydroxide (NaOH)

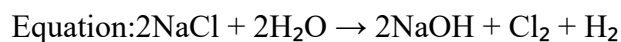
(ii) To decrease pH: Hydrochloric acid (HCl)

(c) (i) When pH decreases from 4 to 2, hydronium ion concentration increases 100 times.

(ii) pH paper colour changes towards red.

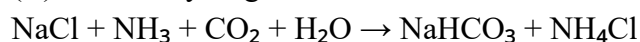
42. (a) X = Chlorine gas (Cl_2), Y = Hydrogen gas (H_2)

(b) Z = Sodium hydroxide (NaOH)

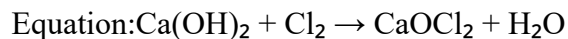


(c) (I) Red litmus turns blue because NaOH is basic.

(II) Sodium hydrogen carbonate is formed.



(III) A = Bleaching powder (CaOCl_2)



43. 1. (a) Sodium chloride + ammonia + water + carbon dioxide

2. (c) Carbon dioxide

3. (a) Tartaric acid

4. (c) It makes the cake soft and spongy

44. 1. (b) Baking soda

2. (c) Carbon dioxide

3. (d) The powder can be used as an antacid

4. (b) Na_2CO_3 , CO_2 and H_2O

45. 1. (c) Calcium carbonate

2. (a) Acid reacted with calcium carbonate

3. (c) Turn lime water milky

4. (b) Calcium sulphate

46. 1. Bleaching powder (CaOCl_2)

2. Preparation reaction: $\text{Ca(OH)}_2 + \text{Cl}_2 \rightarrow \text{CaOCl}_2 + \text{H}_2\text{O}$

3. It is added to drinking water to kill germs and disinfect water.

4. It should be stored in airtight containers because it reacts with moisture and loses its effectiveness.

5. No, baking soda cannot replace bleaching powder because baking soda is not a disinfectant.

6. One other use: Bleaching cotton, linen and paper pulp.



Class 10th
Phodenge 🔥

